

# Courier Management System (CMS) – Comprehensive Project Documentation Report

## Team Details :

Project Title: Courier Management System (CMS)

Program: Salesforce Administrator with AI Agentforce (TN Skills / SmartBridge – Vetri Thiran Payarchi Thittam)

Institution: Mazharul Uloom College (Autonomous), Ambur, Tamil Nadu

Submission Date: September 30, 2026

| S.No | Name                       | Role        |
|------|----------------------------|-------------|
| 1    | P. Mohammed Faiz           | Team Lead   |
| 2    | Sayed Muhammed Mohathaseem | Team Member |
| 3    | Mohammed Sufiyan A         | Team Member |
| 4    | T. Mohammed Azhan          | Team Member |
| 5    | M. G. Maaz                 | Team Member |

## 1. Executive Summary & Project Overview

The Courier Management System (CMS) replaces registers, spreadsheets and phone follow-ups with one Salesforce Lightning application that books shipments, assigns agents, tracks parcel movement, bills customers and reports performance across branches.

### 1.1 Project Metadata

| Attribute       | Detail                                                                                                |
|-----------------|-------------------------------------------------------------------------------------------------------|
| Project name    | Courier Management System (CMS)                                                                       |
| Program track   | Salesforce Administrator with AI Agentforce (TN Skills / SmartBridge – Vetri Thiran Payarchi Thittam) |
| Institution     | Mazharul Uloom College (Autonomous), Ambur, Tamil Nadu                                                |
| Submission date | September 30, 2026                                                                                    |
| Environment     | Salesforce Developer Edition, Lightning Experience                                                    |

| Attribute         | Detail                                                                        |
|-------------------|-------------------------------------------------------------------------------|
| Team lead         | P. Mohammed Faiz                                                              |
| Team members      | Sayed Muhammed Mohathaseem, Mohammed Sufiyan A, T. Mohammed Azhan, M. G. Maaz |
| Delivery approach | Agile, sprint-based (Epics → User Stories → Story Points)                     |
| Custom objects    | 6 (Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice)      |
| Automation        | 9 validation rules, 1 record-triggered flow                                   |
| Analytics         | 3 custom report types, 3 reports, 1 dashboard with 3 components               |

## 1.2 Business Overview of the Logistics and Courier Domain

A courier business earns revenue by moving parcels between a sender and a receiver within a promised time. Every parcel passes through the same lifecycle: a customer books it, a branch accepts it, a delivery agent carries it, status events are recorded along the route, and the customer is invoiced. Each hand-off is a point where information can be lost.

Courier companies typically operate several branches, each with its own manager, agents and customers. Managers need a consolidated view of volume, delays and revenue, while customers expect to know where a parcel is without calling the branch. A domain with many small, time-sensitive transactions and many people touching each record rewards a system that captures every event once and makes it visible to everyone who needs it.

## 1.3 Core Problems Addressed

- **Delayed delivery tracking.** Status is recorded on paper or in spreadsheets and reaches the customer late, or never.
- **Manual agent assignment.** Dispatchers assign parcels by memory or phone, which causes scheduling conflicts and uneven workloads.
- **Lack of customer visibility.** Customers have no single place to check parcel progress, so support staff answer repeated status calls.
- **Fragmented billing.** Shipment, payment and invoice data live in separate files, so invoices are late, duplicated or missing.
- **No performance analytics.** Managers cannot see delivery performance by agent or revenue by branch without hours of manual consolidation.

## 1.4 Solution Summary

CMS is built entirely with declarative Salesforce features, with no Apex code. Six custom objects hold customers, shipments, agents, branches, tracking events and invoices, linked by lookup and master-detail relationships. Validation rules block bad data at entry, and a record-triggered flow copies each new tracking event onto its parent shipment, so shipment status stays current without manual edits.

Role-based security gives Courier Admins full control, Branch Managers branch-level control, Delivery Agents access to their own work, and Customer Support access to customer and tracking

data. A Lightning app groups all modules, and a dashboard turns shipment, agent and revenue data into charts that update on refresh.

| Business outcome         | How CMS delivers it                                                                |
|--------------------------|------------------------------------------------------------------------------------|
| Faster status visibility | Delivery Update records drive shipment status through a flow                       |
| Cleaner data             | Validation rules on Customer, Shipment, Delivery Agent, Branch and Delivery Update |
| Controlled access        | Four profiles and a four-level role hierarchy                                      |
| Managerial insight       | Reports and a three-component dashboard                                            |
| Multi-branch readiness   | Branch and Delivery Agent objects with branch-level grouping                       |

## 2. Problem Statement, Objectives & Scope

Traditional courier operations lose time and money because tracking, assignment, billing and reporting are handled by hand in disconnected tools. CMS is scoped to fix those four areas inside one Salesforce org, and it deliberately leaves out hardware, mobile and external integrations.

### 2.1 Customer Problem Statements

| ID   | I am (persona) | I'm trying to                | But                 | Because                        | Which makes me feel |
|------|----------------|------------------------------|---------------------|--------------------------------|---------------------|
| PS-1 | Branch Manager | manage shipments efficiently | tracking is manual  | systems are disconnected       | stressed            |
| PS-2 | Customer       | track parcels                | updates are delayed | there is no centralized system | uncertain           |

### 2.2 Operational Bottlenecks in Traditional Courier Tracking

| #   | Bottleneck                   | Typical cause                                                           | Consequence                                          |
|-----|------------------------------|-------------------------------------------------------------------------|------------------------------------------------------|
| B-1 | Status updates recorded late | Agents report by phone or at end of shift; clerks re-key into registers | Customers see stale status; managers act on old data |
| B-2 | Manual agent assignment      | Dispatcher decides from memory or chat messages                         | Double-booking, idle agents, missed pickups          |
| B-3 | No single shipment record    | Details spread across registers, spreadsheets and paper slips           | Duplicate entry, transcription errors, lost history  |

| #   | Bottleneck                    | Typical cause                                                          | Consequence                                           |
|-----|-------------------------------|------------------------------------------------------------------------|-------------------------------------------------------|
| B-4 | Unvalidated data entry        | Free-text phone numbers, zero or negative weights, past delivery dates | Failed callbacks, wrong pricing, unreliable estimates |
| B-5 | No customer-facing visibility | Customers must call the branch for status                              | High support load and repeated queries                |
| B-6 | Fragmented billing            | Invoices prepared separately from shipment data                        | Delayed, missing or disputed invoices                 |
| B-7 | No performance analytics      | Data is not structured for aggregation                                 | Managers cannot compare agents or branches            |
| B-8 | Weak access control           | Shared files and open registers                                        | Sensitive customer and billing data exposed           |

### 2.3 Business Impact Matrix

| Impact area            | Current-state effect                                                                                      | Severity | CMS response                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|
| Customer satisfaction  | Delayed updates, no visibility, repeated status calls, uncertainty about delivery                         | High     | Delivery Update log drives shipment status automatically; Customer Support can see tracking data in one place |
| Operational efficiency | Manual assignment, re-keying, scheduling conflicts, coordination gaps between branches                    | High     | Assigned Agent lookup, validation rules, automated status flow, branch-linked agents                          |
| Revenue loss           | Unbilled or late-billed shipments, disputes from wrong weight or price, lost customers after poor service | High     | Invoice linked to Shipment with payment status; price and weight validation                                   |
| Management insight     | Decisions made on anecdote rather than data                                                               | Medium   | Reports and Courier Management Dashboard                                                                      |
| Data security          | Uncontrolled access to customer and billing records                                                       | Medium   | Profiles, roles and object-level permissions                                                                  |

## 2.4 Project Objectives

1. Centralize customer, shipment, agent, branch, tracking and invoice data in one Salesforce org.
2. Enforce data quality at entry through validation rules on the fields that cause most rework.
3. Update shipment status automatically from delivery events, with no manual edit of the parent shipment.
4. Restrict each user group to the data and actions its job needs, through profiles and roles.
5. Provide managers with reports and a dashboard for shipment status, agent performance and branch revenue.
6. Keep the solution declarative, so administrators can maintain it without code.

## 2.5 Scope Boundaries

| In scope                                                                        | Out of scope (future scope)                               |
|---------------------------------------------------------------------------------|-----------------------------------------------------------|
| Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice objects | Mobile tracking app                                       |
| Validation rules and the record-triggered shipment status flow                  | GPS or live location tracking                             |
| Lightning app, custom tabs and page layouts                                     | SMS or email notifications to customers                   |
| Profiles, roles and users for four user groups                                  | AI-based delivery route optimization                      |
| Custom report types, reports and one dashboard                                  | Customer self-service portal or Experience Cloud site     |
| Functional testing with screenshot evidence                                     | Payment gateway integration and online payment collection |
| User guide and deployment checklist                                             | Integration with external ERP or accounting systems       |

### 2.5.1 Assumptions and Constraints

- The system runs in a Salesforce Developer Edition org with limited user licenses, so only a small number of full Salesforce licenses can be active at once.
- Users access the system through Lightning Experience in a web browser with internet connectivity.
- Notifications to customers are not automated in this release; Customer Support relays status when asked.
- Invoice amounts are entered or copied by staff; there is no automatic tax or discount engine.

## 3. System Requirements Specification (SRS)

CMS must satisfy six functional requirements and four non-functional requirements, each traceable to a build artifact and a test. Priority follows MoSCoW (Must, Should, Could).

### 3.1 Functional Requirements

| ID | Requirement | Description | Priority | Realized by |
|----|-------------|-------------|----------|-------------|
|----|-------------|-------------|----------|-------------|

| ID    | Requirement                   | Description                                                                                                                                                                                                                                                                                                 | Priority | Realized by                                |
|-------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------|
| FR-01 | Customer management           | Create, view, edit and deactivate customer records with name, email, phone, address, city, state, pincode, customer type (Regular or Business) and an Is Active flag. Phone must be exactly 10 digits and pincode 6 digits.                                                                                 | Must     | Customer__c, validation rules, page layout |
| FR-02 | Shipment booking              | Book a shipment against a customer with source and destination address, weight, price, shipment type (Domestic or International), expected delivery date and status. Weight and price must be greater than zero, and the expected date cannot be in the past. Tracking numbers are generated automatically. | Must     | Shipment__c, validation rules, auto-number |
| FR-03 | Agent assignment and tracking | Maintain delivery agents with branch, status (Active, Inactive, On Leave) and joining date; assign one agent to each shipment through the Assigned Agent                                                                                                                                                    | Must     | Delivery_Agent__c, Branch__c, lookups      |

| ID    | Requirement              | Description                                                                                                                                                                                                                 | Priority | Realized by                                                |
|-------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------|
|       |                          | lookup; list each agent's shipments.                                                                                                                                                                                        |          |                                                            |
| FR-04 | Automated status updates | Each Delivery Update record (In Transit, Out for Delivery, Delivered, Delayed) is timestamped and linked to its shipment; the shipment status changes automatically without manual edit. A Delayed update requires remarks. | Must     | Delivery_Update_<br>_c,<br>Update_Shipment<br>_Status_Flow |
| FR-05 | Invoicing                | Generate an invoice per shipment with amount, invoice date and payment status (Paid, Unpaid, Pending). Invoice numbers are generated automatically.                                                                         | Must     | Invoice__c                                                 |
| FR-06 | Reporting and dashboards | Provide reports on shipment status, agent performance and revenue by branch, and a dashboard that visualizes them for managers.                                                                                             | Should   | Report types,<br>reports,<br>dashboard                     |

### 3.2 FR Acceptance Criteria

| ID    | Acceptance criterion                                                                            |
|-------|-------------------------------------------------------------------------------------------------|
| FR-01 | Saving a customer with a phone number that is not exactly 10 digits is blocked with an error on |

| ID    | Acceptance criterion                                                                                                                 |
|-------|--------------------------------------------------------------------------------------------------------------------------------------|
|       | the Phone field.                                                                                                                     |
| FR-02 | Saving a shipment with weight of 0 or less is blocked; a saved shipment receives a unique auto-numbered tracking ID.                 |
| FR-03 | A shipment can be saved with an Assigned Agent, and the agent record lists that shipment in a related list.                          |
| FR-04 | Creating a Delivery Update with status In Transit changes the parent shipment from Booked to In Transit within the same transaction. |
| FR-05 | An invoice created against SHIP-0001 appears in that shipment's Invoices related list.                                               |
| FR-06 | Refreshing the Courier Management Dashboard renders all three components with current data.                                          |

### 3.3 Non-Functional Requirements

| ID     | Category              | Requirement                                                                                                                                                                                                                             | Measure / how verified                                               |
|--------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| NFR-01 | Data integrity        | Invalid data must be rejected at save, parent-child relationships must remain consistent, and required fields must always be populated. Delivery Updates are master-detail children of Shipment, so they cannot exist without a parent. | Validation rule tests (TC-1, TC-2); relationship checks (TC-3, TC-4) |
| NFR-02 | Role-based security   | Every user is assigned one profile and one role. Users see and change only the objects and records their job requires; delivery agents cannot delete records or edit branches.                                                          | Profile object permissions; role hierarchy; user login checks        |
| NFR-03 | Platform availability | The system relies on the Salesforce cloud,                                                                                                                                                                                              | Salesforce platform service level; login test                        |

| ID     | Category              | Requirement                                                                                                                              | Measure / how verified                          |
|--------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
|        |                       | giving 24/7 browser access without local servers, installation or maintenance windows managed by the team.                               | from a fresh browser                            |
| NFR-04 | Dashboard scalability | Dashboards must remain readable as branches, agents and shipments grow, by using grouped summary reports rather than row-level listings. | Dashboard refresh (TC-5); grouped report design |

### 3.4 Additional Quality Attributes from the Base Design

| Attribute   | Description                                                                          |
|-------------|--------------------------------------------------------------------------------------|
| Usability   | Lightning interface with a single app, one tab per object and sectioned page layouts |
| Reliability | Automation is declarative and runs in the same transaction as the triggering record  |
| Performance | The status flow runs after save and updates one parent record                        |
| Scalability | New branches and agents are added as records with no configuration change            |

### 3.5 Technology Stack

| Layer          | Technology                                    |
|----------------|-----------------------------------------------|
| User interface | Salesforce Lightning Experience               |
| Business logic | Salesforce Flow                               |
| Data           | Salesforce custom objects                     |
| Automation     | Validation rules and record-triggered flow    |
| Security       | Profiles, roles, object and field permissions |
| Reporting      | Custom report types, reports and dashboards   |

### 3.6 Data Flow

Customer input → Shipment creation → Validation → Agent assignment → Delivery update → Invoice → Reports

## 4. Data Model & System Architecture

The data model has six custom objects joined by five relationships: four lookups and one master-detail. Shipment is the central object; every other object connects to it directly or through the Delivery Agent.

### 4.1 Architecture Components

| Layer        | Components                                                                                       |
|--------------|--------------------------------------------------------------------------------------------------|
| Data         | Customer__c, Shipment__c, Delivery_Agent__c, Branch__c, Delivery_Update__c, Invoice__c           |
| Logic        | 9 validation rules, Update_Shipment_Status_Flow                                                  |
| Presentation | Courier Management System Lightning app, 6 custom tabs plus Reports and Dashboards, page layouts |
| Security     | 4 profiles, 4 roles, users                                                                       |
| Analytics    | 3 custom report types, 3 reports, Courier Management Dashboard                                   |

### 4.2 Data Dictionary

Lengths and precisions are the Salesforce defaults accepted during the build unless stated otherwise. Every object also carries the standard system fields (Id, Owner, Created By, Created Date, Last Modified By, Last Modified Date), which are not repeated below. Master-detail children have no Owner field; they inherit the parent's owner and sharing.

#### 4.2.1 Customer\_\_c (Customers)

Stores the person or business that books shipments. Record name is a text field; Allow Reports, Track Field History, Allow Search and Notes & Attachments are enabled.

| Field Label   | API Name   | Data Type | Length / Precision | Required | Description                                    |
|---------------|------------|-----------|--------------------|----------|------------------------------------------------|
| Customer Name | Name       | Text      | 80                 | Y        | Full name of the customer                      |
| Email         | Email__c   | Email     | 80                 | Y        | Customer email address                         |
| Phone         | Phone__c   | Phone     | 40                 | Y        | Contact number; validated as exactly 10 digits |
| Address       | Address__c | Text Area | 255                | Y        | Customer street address                        |

| Field Label   | API Name         | Data Type | Length / Precision        | Required | Description                              |
|---------------|------------------|-----------|---------------------------|----------|------------------------------------------|
| Customer Type | Customer_Type__c | Picklist  | Values: Regular, Business | Y        | Classifies the customer                  |
| City          | City__c          | Text      | 255                       | N        | Customer city                            |
| State         | State__c         | Text      | 255                       | N        | Customer state                           |
| Pincode       | Pincode__c       | Text      | 255                       | N        | Postal code; validated as 6 digits       |
| Is Active     | Is_Active__c     | Checkbox  | Default unchecked         | N        | Whether the customer is currently active |

#### 4.2.2 Shipment\_\_c (Shipments)

The consignment record. Record name is an auto-number in the format SHP-{00000}. Allow Reports and Track Field History are enabled.

| Field Label         | API Name               | Data Type        | Length / Precision    | Required | Description                             |
|---------------------|------------------------|------------------|-----------------------|----------|-----------------------------------------|
| Tracking Number     | Name                   | Auto Number      | SHP-{00000}           | Y        | Unique shipment identifier              |
| Customer            | Customer__c            | Lookup(Customer) | Lookup to Customer__c | Y        | Customer who booked the shipment        |
| Source Address      | Source_Address__c      | Text Area        | 255                   | Y        | Pickup location                         |
| Destination Address | Destination_Address__c | Text Area        | 255                   | Y        | Delivery location                       |
| Weight              | Weight__c              | Number           | 18,2                  | Y        | Parcel weight; must be greater than 0   |
| Price               | Price__c               | Currency         | 18,2                  | Y        | Shipping charge; must be greater than 0 |
| Status              | Status__c              | Picklist         | Values: Booked, In    | Y        | Current shipment                        |

| Field Label    | API Name          | Data Type              | Length / Precision                              | Required | Description                                    |
|----------------|-------------------|------------------------|-------------------------------------------------|----------|------------------------------------------------|
|                |                   |                        | Transit, Out for Delivery, Delivered, Cancelled |          | status                                         |
| Expected Date  | Expected_Date__c  | Date                   | Date                                            | Y        | Estimated delivery date; cannot be in the past |
| Assigned Agent | Assigned_Agent__c | Lookup(Delivery Agent) | Lookup to Delivery_Agent__c                     | N        | Delivery agent responsible                     |
| Shipment Type  | Shipment_Type__c  | Picklist               | Values: Domestic, International                 | Y        | Delivery scope                                 |

#### 4.2.3 Delivery\_Agent\_\_c (Delivery Agents)

Logistics personnel who carry shipments. Allow Reports and Track Field History are enabled.

| Field Label  | API Name        | Data Type      | Length / Precision                 | Required | Description                 |
|--------------|-----------------|----------------|------------------------------------|----------|-----------------------------|
| Agent Name   | Name            | Text           | 80                                 | Y        | Full name of the agent      |
| Email        | Email__c        | Email          | 80                                 | Y        | Agent email; must contain @ |
| Phone        | Phone__c        | Phone          | 40                                 | Y        | Agent contact number        |
| Branch       | Branch__c       | Lookup(Branch) | Lookup to Branch__c                | Y        | Branch the agent works from |
| Status       | Status__c       | Picklist       | Values: Active, Inactive, On Leave | Y        | Availability of the agent   |
| Joining Date | Joining_Date__c | Date           | Date                               | N        | Date the agent joined       |

#### 4.2.4 Branch\_\_c (Branches)

Regional hub or branch office. Allow Reports is enabled; field history is not tracked.

| Field Label  | API Name        | Data Type | Length / Precision | Required | Description                       |
|--------------|-----------------|-----------|--------------------|----------|-----------------------------------|
| Branch Name  | Name            | Text      | 80                 | Y        | Name of the branch                |
| Branch Code  | Branch_Code__c  | Text      | 255                | Y        | Short unique code; mandatory      |
| City         | City__c         | Text      | 255                | Y        | City where the branch is located  |
| State        | State__c        | Text      | 255                | Y        | State where the branch is located |
| Manager Name | Manager_Name__c | Text      | 255                | N        | Name of the branch manager        |

#### 4.2.5 Delivery\_Update\_\_c (Delivery Updates)

An event log entry for a shipment. Record name is an auto-number in the format UPD-{0000}. Allow Reports and Track Field History are enabled.

| Field Label   | API Name       | Data Type               | Length / Precision                                       | Required | Description                            |
|---------------|----------------|-------------------------|----------------------------------------------------------|----------|----------------------------------------|
| Update Number | Name           | Auto Number             | UPD-{0000}                                               | Y        | Unique update identifier               |
| Shipment      | Shipment__c    | Master-Detail(Shipment) | Master-detail to Shipment__c                             | Y        | Parent shipment                        |
| Status        | Status__c      | Picklist                | Values: In Transit, Out for Delivery, Delivered, Delayed | Y        | Status reported by this event          |
| Update Date   | Update_Date__c | Date/Time               | Date/Time                                                | Y        | When the event occurred                |
| Remarks       | Remarks__c     | Text                    | 255                                                      | N        | Notes; required when status is Delayed |

#### 4.2.6 Invoice\_\_c (Invoices)

Billing record for a shipment. Record name is an auto-number in the format INV-{0000}. Allow Reports and Track Field History are enabled.

| Field Label    | API Name          | Data Type        | Length / Precision            | Required | Description                 |
|----------------|-------------------|------------------|-------------------------------|----------|-----------------------------|
| Invoice Number | Name              | Auto Number      | INV-{0000}                    | Y        | Unique invoice identifier   |
| Shipment       | Shipment__c       | Lookup(Shipment) | Lookup to Shipment__c         | Y        | Shipment being billed       |
| Amount         | Amount__c         | Currency         | 18,2                          | Y        | Invoice amount              |
| Invoice Date   | Invoice_Date__c   | Date             | Date                          | Y        | Date the invoice was raised |
| Payment Status | Payment_Status__c | Picklist         | Values: Paid, Unpaid, Pending | Y        | Payment state               |

#### 4.3 Object Relationships

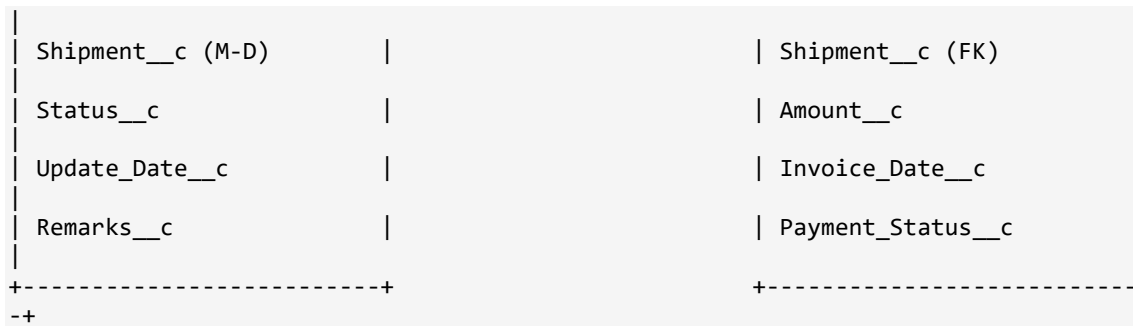
| Child object       | Relationship field | Parent object     | Type          | Behavior                                                                            |
|--------------------|--------------------|-------------------|---------------|-------------------------------------------------------------------------------------|
| Shipment__c        | Customer__c        | Customer__c       | Lookup        | Each shipment belongs to one customer; a customer can have many shipments           |
| Shipment__c        | Assigned_Agent__c  | Delivery_Agent__c | Lookup        | Each shipment has at most one agent; an agent can carry many shipments              |
| Delivery_Agent__c  | Branch__c          | Branch__c         | Lookup        | Each agent belongs to one branch; a branch has many agents                          |
| Delivery_Update__c | Shipment__c        | Shipment__c       | Master-Detail | Updates are deleted with their shipment and roll up to it; inherit sharing from the |

| Child object | Relationship field | Parent object | Type   | Behavior                                                                             |
|--------------|--------------------|---------------|--------|--------------------------------------------------------------------------------------|
|              |                    |               |        | shipment                                                                             |
| Invoice__c   | Shipment__c        | Shipment__c   | Lookup | An invoice references one shipment; a shipment can have more than one invoice record |

A lookup can be left empty or the parent deleted independently, so Shipment survives if an Assigned Agent is removed. The master-detail link makes Delivery Update depend on Shipment for ownership, sharing and deletion, which protects the tracking history from orphan records.

#### 4.4 Entity-Relationship Diagram (text form)





Reading the diagram: arrows point from child to parent. Customer, Delivery Agent, Delivery Update and Invoice all reference Shipment or feed into it, and Branch sits above Delivery Agent, so a report can walk from Invoice to Shipment to Agent to Branch.

| Relationship               | Cardinality | Delete behavior                                                                                             |
|----------------------------|-------------|-------------------------------------------------------------------------------------------------------------|
| Customer → Shipment        | 1 : N       | Lookup; shipments are kept if the customer is deleted only when the lookup is configured to clear the value |
| Branch → Delivery Agent    | 1 : N       | Lookup; agents are kept if the branch is deleted only when the lookup is configured to clear the value      |
| Delivery Agent → Shipment  | 1 : N       | Lookup; shipment stays when the agent is removed                                                            |
| Shipment → Delivery Update | 1 : N       | Master-detail; updates are deleted with the shipment                                                        |
| Shipment → Invoice         | 1 : N       | Lookup; invoice is kept unless the lookup is configured to restrict deletion                                |

## 5. Business Logic & Automation Specifications

CMS enforces data quality with nine validation rules and keeps shipment status current with one record-triggered flow. Validation rules run when a record is saved and block the save while the formula returns TRUE.

### 5.1 Validation Rules Summary

| # | Object      | Rule name      | Error condition formula            | Error message                   | Error location |
|---|-------------|----------------|------------------------------------|---------------------------------|----------------|
| 1 | Customer__c | Email_Required | ISBLANK(Email__c)                  | Email is required.              | Email          |
| 2 | Customer__c | Validate_Phone | NOT(REGEX(Phone__c, ^[0-9]{10}\$)) | Phone number must be 10 digits. | Phone          |

| # | Object             | Rule name                  | Error condition formula                                   | Error message                                 | Error location |
|---|--------------------|----------------------------|-----------------------------------------------------------|-----------------------------------------------|----------------|
|   |                    | e_Number                   | "^[0-9]{10}\$")                                           | exactly 10 digits.                            |                |
| 3 | Customer__c        | Validate_Pincode           | LEN(Pincode__c) <> 6                                      | Pincode must be 6 digits.                     | Pincode        |
| 4 | Shipment__c        | Validate_Weight            | Weight__c <= 0                                            | Shipment weight must be greater than 0.       | Weight         |
| 5 | Shipment__c        | Validate_Price             | Price__c <= 0                                             | Price must be greater than zero.              | Price          |
| 6 | Shipment__c        | Validate_Expected_Date     | Expected_Date__c < TODAY()                                | Expected delivery date cannot be in the past. | Expected Date  |
| 7 | Delivery_Agent__c  | Validate_Agent_Email       | NOT(CONTAINS(Email__c, "@"))                              | Enter a valid email address.                  | Email          |
| 8 | Branch__c          | Branch_Code_Required       | ISBLANK(Branch_Code__c)                                   | Branch Code is mandatory.                     | Branch Code    |
| 9 | Delivery_Update__c | Remarks_Required_For_Delay | AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c)) | Remarks are required when status is Delayed.  | Remarks        |

## 5.2 Customer Phone Validation

- **Rule name:** Validate\_Phone\_Number, object Customer\_\_c, Active.
- **Formula:** NOT(REGEX(Phone\_\_c, "^[0-9]{10}\$"))
- **Error message:** Phone number must be exactly 10 digits.
- **Error location:** Field, Phone.

The regular expression `^[0-9]{10}$` matches a value that starts, contains exactly ten characters from 0 to 9, and ends. Anything else, such as nine digits, eleven digits, spaces, dashes, a plus sign or letters, makes REGEX return FALSE, NOT turns that into TRUE, and the save is blocked. A stricter check than testing length alone, it also rejects ten characters that are not all digits. If the org's Phone field auto-formats input, enter digits only.

Steps to create: Setup → Object Manager → Customer\_\_c → Validation Rules → New → enter the rule name, tick Active, paste the formula, click Check Syntax, enter the error message, set the location to the Phone field, and Save.

### 5.3 Shipment Weight Validation

- **Rule name:** Validate\_Weight, object Shipment\_\_c, Active.
- **Formula:** Weight\_\_c <= 0
- **Error message:** Shipment weight must be greater than 0.
- **Error location:** Field, Weight.

The rule returns TRUE for weights of zero or below, so a shipment cannot be booked without a positive weight. Weight drives pricing, so a zero weight would produce a wrong charge and a wrong invoice.

### 5.4 Validation Design Notes

- **Expected date and flow updates.** Validation rules also run when a flow updates a record. Because Expected\_Date\_\_c < TODAY() is evaluated on every save, a shipment whose expected date has passed cannot be updated by the status flow until the date is corrected. If this appears in production, change the formula to AND(ISNEW(), Expected\_Date\_\_c < TODAY()) or ISCHANGED(Expected\_Date\_\_c) combined with the date test, so old shipments can still change status.
- **Blank values.** Fields marked Required already block blanks; the formulas above therefore test only the condition that is not covered by Required.
- **Error location.** Field-level messages appear under the field, which helps agents fix the exact input.

### 5.5 Record-Triggered Flow: Update\_Shipment\_Status\_Flow

| Property              | Value                                                            |
|-----------------------|------------------------------------------------------------------|
| Flow label / API name | Update_Shipment_Status_Flow                                      |
| Flow type             | Record-Triggered Flow (after save)                               |
| Trigger object        | Delivery_Update__c                                               |
| Trigger               | A record is created or updated                                   |
| Entry condition       | Status__c Equals In Transit                                      |
| Run frequency         | Only when a record is updated to meet the condition requirements |
| Optimize for          | Actions and Related Records                                      |
| Element               | Update Records: Update_Shipment_Status                           |
| Target                | Shipment__c where Id = {!\$Record.Shipment__c}                   |
| Field change          | Status__c = In Transit                                           |
| Status                | Active                                                           |

#### Logic flow

```
[Start] Delivery_Update__c created or updated
|
v
```

```

[Entry condition] Status__c = "In Transit" ?
|-- No --> Flow ends, Shipment unchanged
|-- Yes -->
    v
[Update Records] Update_Shipment_Status
    Find:    Shipment__c WHERE Id = {!$Record.Shipment__c}
    Set:     Status__c = "In Transit"
    v
[End] Shipment shows In Transit

```

### Step-by-step build

1. Setup → Quick Find → Flows → New Flow → Record-Triggered Flow → Create.
2. On the Start element, choose Object = Delivery\_Update\_\_c.
3. Set Trigger the Flow When to A record is created or updated.
4. Under Set Entry Conditions, choose Condition Requirements = All Conditions Are Met (AND), with Field Status\_\_c, Operator Equals, Value In Transit.
5. Set When to Run the Flow for Updated Records to Only when a record is updated to meet the condition requirements, so the flow does not fire again on edits that leave the status unchanged.
6. Under Optimize the Flow for, choose Actions and Related Records, then click Done.
7. Click + under Start and add an Update Records element labeled Update\_Shipment\_Status.
8. Choose Specify conditions to identify records, and set the object to Shipment\_\_c.
9. Add the condition Id Equals {!\$Record.Shipment\_\_c}, so only the parent shipment is selected.
10. Under Set Field Values, add Status\_\_c = In Transit. Click Done.
11. Click Save, enter the label Update\_Shipment\_Status\_Flow, and accept the generated API name.
12. Click Activate and confirm the status shows Active.

### Runtime behavior

| Scenario                                                        | Result                                                                           |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
| New Delivery Update with status In Transit                      | Parent shipment status becomes In Transit                                        |
| Existing update edited from Delayed to In Transit               | Flow fires because the record now meets the condition; parent becomes In Transit |
| Update saved with status Out for Delivery, Delivered or Delayed | Flow does not run; parent shipment status is unchanged                           |
| Shipment already In Transit                                     | Update runs but writes the same value; no visible change                         |
| Parent shipment fails a validation rule on save                 | Flow fault; the Delivery Update save is rolled back and the error is shown       |
| User has no edit access to the shipment                         | Flow runs in system context and still updates the parent                         |

## 5.6 Automation Scope and Recommended Extension

As specified, the flow moves shipments from Booked to In Transit only. Later statuses are still set by staff on the shipment. The base build note describes copying each update's status onto the shipment, which would also cover Out for Delivery and Delivered; that extension needs a Decision element with one outcome per status, because Delayed exists on Delivery Update but is not a value of the Shipment Status picklist and would otherwise cause a save error. The extension is listed in Section 10 under maintenance recommendations.

## 6. Security & Access Control Model

CMS uses three layers: profiles decide what a user can do, roles decide which records a user can see, and sharing defaults set the baseline for record access. Profiles control permissions; roles control data visibility.

### 6.1 Organization-Wide Defaults (OWD)

The base build configures profiles and roles; the sharing defaults below are the design baseline for the org and should be confirmed in Setup → Sharing Settings.

| Object             | Default internal access | Rationale                                                                                                      |
|--------------------|-------------------------|----------------------------------------------------------------------------------------------------------------|
| Customer__c        | Private                 | Contains personal contact details; visible to the owner and to managers above the owner in the role hierarchy  |
| Shipment__c        | Private                 | Consignments belong to the branch that booked them; managers see subordinates' shipments through the hierarchy |
| Delivery_Agent__c  | Public Read Only        | Agent names and branches are needed for assignment by all staff, but only managers edit them                   |
| Branch__c          | Public Read Only        | Reference data used across the app                                                                             |
| Delivery_Update__c | Controlled by Parent    | Master-detail child; access follows the parent shipment                                                        |
| Invoice__c         | Private                 | Financial data restricted to the owner and managers                                                            |

With Private defaults, the Grant Access Using Hierarchies option must stay enabled so a Branch Manager sees records owned by roles beneath.

### 6.2 Role Hierarchy

CEO (standard top role)

|

```

+-- Courier Admin
  |
  +-- Branch Manager
    |
    +-- Delivery Agent
      |
      +-- Customer Support

```

| Role             | Reports to     | Data visibility purpose                                                                          |
|------------------|----------------|--------------------------------------------------------------------------------------------------|
| Courier Admin    | CEO            | Highest level access to all data; owns configuration                                             |
| Branch Manager   | Courier Admin  | Manages branch-level operations; sees records owned by Delivery Agent and Customer Support users |
| Delivery Agent   | Branch Manager | Works on assigned shipments and tracking updates                                                 |
| Customer Support | Branch Manager | Limited access to customer and shipment tracking data                                            |

Delivery Agent and Customer Support sit at the same level, so neither sees the other's private records; both are visible to the Branch Manager above them.

### 6.3 Profile Definitions

| Profile          | Cloned from            | Purpose                                                                               | Default app               | Visible tabs                                                                                     |
|------------------|------------------------|---------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------|
| Courier Admin    | System Administrator   | Full system control: Modify All Data, Customize Application, full CRUD on all objects | Courier Management System | All                                                                                              |
| Branch Manager   | Standard Platform User | Manage shipments and agents                                                           | Courier Management System | Customers, Shipments, Delivery Agents, Branches, Delivery Updates, Invoices, Reports, Dashboards |
| Delivery Agent   | Standard Platform User | Update delivery status                                                                | Courier Management System | Shipments, Delivery Updates                                                                      |
| Customer Support | Standard User          | Handle customers and tracking                                                         | Courier Management System | Customers, Shipments, Delivery Updates                                                           |

## 6.4 Object-Level Permissions

C = Create, R = Read, E = Edit, D = Delete. A dash means no access.

| Object             | Courier Admin             | Branch Manager | Delivery Agent | Customer Support |
|--------------------|---------------------------|----------------|----------------|------------------|
| Customer__c        | CRED +<br>View/Modify All | C, R, E        | R              | R, E             |
| Shipment__c        | CRED +<br>View/Modify All | C, R, E        | R, E           | R, E             |
| Delivery_Agent__c  | CRED +<br>View/Modify All | R, E           | –              | –                |
| Branch__c          | CRED +<br>View/Modify All | R, E           | –              | –                |
| Delivery_Update__c | CRED +<br>View/Modify All | C, R, E        | C, R, E        | R                |
| Invoice__c         | CRED +<br>View/Modify All | R              | R              | R                |

Delivery Agents have no delete permission and cannot edit branches. Only the Courier Admin can delete records.

## 6.5 Session and Password Settings

| Setting                 | Branch Manager        | Delivery Agent | Customer Support | Courier Admin |
|-------------------------|-----------------------|----------------|------------------|---------------|
| Session timeout         | 2 hours of inactivity | Org default    | 2 hours          | Org default   |
| Minimum password length | 8                     | Org default    | 8                | Org default   |
| Password expiry         | Never expires         | Org default    | Org default      | Org default   |

## 6.6 Field-Level Security and Permission Sets

Field-level security (FLS) is set per profile on each field and takes precedence over what a page layout shows. The table below is the recommended FLS baseline for fields that carry money or personal data.

| Field                                      | Courier Admin | Branch Manager | Delivery Agent | Customer Support |
|--------------------------------------------|---------------|----------------|----------------|------------------|
| Customer__c.Email__c, Phone__c, Address__c | Edit          | Edit           | Read           | Edit             |
| Shipment__c.Price__c                       | Edit          | Edit           | Read           | Read             |
| Shipment__c.Stat                           | Edit          | Edit           | Edit           | Read             |

| Field                                     | Courier Admin | Branch Manager | Delivery Agent | Customer Support |
|-------------------------------------------|---------------|----------------|----------------|------------------|
| us__c                                     |               |                |                |                  |
| Delivery_Update__c, Status__c, Remarks__c | Edit          | Edit           | Edit           | Read             |
| Invoice__c, Amount__c, Payment_Status__c  | Edit          | Read           | Read           | Read             |

Permission sets extend a profile without cloning it. Recommended sets, not yet created in the build:

| Permission set     | Grants                                                         | Assign to                                   |
|--------------------|----------------------------------------------------------------|---------------------------------------------|
| CMS_Invoice_Editor | Create and Edit on Invoice__c, Edit on Payment_Status__c       | Selected Branch Managers or finance staff   |
| CMS_Report_Builder | Create and Customize Reports, Manage Dashboards in CMS folders | Branch Managers who build their own reports |
| CMS_Data_Exporter  | Export Reports                                                 | Courier Admin only                          |

## 6.7 User Provisioning

| User type        | Role             | Profile          | License in the base build |
|------------------|------------------|------------------|---------------------------|
| Courier Admin    | Courier Admin    | Courier Admin    | Salesforce                |
| Branch Manager   | Branch Manager   | Branch Manager   | Salesforce                |
| Delivery Agent   | Delivery Agent   | Delivery Agent   | Salesforce Platform       |
| Customer Support | Customer Support | Customer Support | Salesforce Platform       |

Each user needs a unique username in email format, a valid email address, one role and one profile. A Developer Edition org has a small number of full Salesforce licenses; if none is free, deactivate an unused user or change the license type.

License compatibility: a profile cloned from a Platform base can be assigned only to a Salesforce Platform user, and a profile cloned from Standard User only to a Salesforce user. If Setup rejects an assignment for Branch Manager or Customer Support, switch the user's license so it matches the profile's base, or re-clone the profile from the matching base.

## 7. User Interface & Lightning App Configuration

Staff work in one Lightning app, Courier Management System, that carries a tab for every object plus Reports and Dashboards. Each object has a sectioned page layout so users see related fields together.

## 7.1 Lightning App Configuration

| Setting                       | Value                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| App name                      | Courier Management System                                                                            |
| Type                          | Lightning app (standard navigation)                                                                  |
| App options                   | Defaults                                                                                             |
| Utility items                 | Defaults (none added)                                                                                |
| Navigation items              | Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice, Reports, Dashboards            |
| Assigned profiles at creation | System Administrator                                                                                 |
| Profiles that must be added   | Courier Admin, Branch Manager, Delivery Agent, Customer Support (App Manager → Edit → User Profiles) |

Create it from Setup → App Manager → New Lightning App: enter the name and an icon, accept defaults for App Options and Utility Items, move the eight items into Selected Items, select profiles, and click Save & Finish.

## 7.2 Custom Tabs

| Tab              | Object             | Created from                            | Tab style |
|------------------|--------------------|-----------------------------------------|-----------|
| Customers        | Customer__c        | Setup → Tabs → Custom Object Tabs → New | Any icon  |
| Shipments        | Shipment__c        | Same path                               | Any icon  |
| Delivery Agents  | Delivery_Agent__c  | Same path                               | Any icon  |
| Branches         | Branch__c          | Same path                               | Any icon  |
| Delivery Updates | Delivery_Update__c | Same path                               | Any icon  |
| Invoices         | Invoice__c         | Same path                               | Any icon  |

Tabs are created with default visibility and then made visible per profile as listed in Section 6. The suggested navigation order follows the shipment lifecycle: Customers, Shipments, Delivery Agents, Branches, Delivery Updates, Invoices, Reports, Dashboards.

## 7.3 Page Layout: Shipment\_\_c

| Section             | Columns | Fields                                                          |
|---------------------|---------|-----------------------------------------------------------------|
| Shipment Details    | 2       | Tracking Number, Customer, Shipment Type, Weight, Price, Status |
| Address Information | 2       | Source Address, Destination                                     |

| Section              | Columns    | Fields                                          |
|----------------------|------------|-------------------------------------------------|
|                      |            | Address                                         |
| Delivery Information | 2          | Expected Date, Assigned Agent                   |
| Related lists        | Full width | Delivery Updates, Invoices, Notes & Attachments |

Required fields carry the red marker: Customer, both addresses, Weight, Price, Status, Expected Date and Shipment Type. Keep Status and Assigned Agent near the top of the page for dispatchers by placing Delivery Information directly after Shipment Details if the team prefers.

#### 7.4 Page Layout: Customer\_\_c

| Section              | Columns    | Fields                                                |
|----------------------|------------|-------------------------------------------------------|
| Customer Information | 2          | Customer Name, Customer Type, Email, Phone, Is Active |
| Address Details      | 2          | Address, City, State, Pincode                         |
| Related lists        | Full width | Shipments, Notes & Attachments                        |

Tab-key order is top-down in both sections, so users can enter a customer without using the mouse.

#### 7.5 Page Layouts for the Remaining Objects

| Object             | Section                | Fields and related lists                                                           |
|--------------------|------------------------|------------------------------------------------------------------------------------|
| Delivery_Agent__c  | Agent Details          | Agent Name, Email, Phone, Branch, Status, Joining Date                             |
| Delivery_Agent__c  | Additional Information | Related lists: Shipments, Delivery Updates                                         |
| Branch__c          | Branch Information     | Branch Name, Branch Code, City, State, Manager Name; related list: Delivery Agents |
| Delivery_Update__c | Update Information     | Update Number, Shipment, Status, Update Date, Remarks                              |
| Invoice__c         | Invoice Details        | Invoice Number, Shipment, Amount, Invoice Date, Payment Status                     |

#### 7.6 Layout Build Steps

1. Setup → Object Manager → select the object → Page Layouts → open the default layout.
2. Drag Section from the palette onto the canvas, name it, choose a 2-column layout and Top-Down tab order, then click OK.
3. Drag each field from the palette into its section.
4. Add or remove related lists in the Related Lists area.

5. Click Save.

## 8. System Testing & Test Execution Results

All five functional test cases passed. Each one was captured as a Salesforce screenshot that proves record creation, validation enforcement, flow automation, invoice linking or dashboard rendering, as stated in the project's functional testing summary.

### 8.1 Test Approach

| Item           | Detail                                                        |
|----------------|---------------------------------------------------------------|
| Test type      | Manual functional testing in the Salesforce Lightning UI      |
| Environment    | Salesforce Developer Edition, Courier Management System app   |
| Tester profile | Courier Admin (full access)                                   |
| Evidence       | One screenshot artifact per test case (SS-1 to SS-5)          |
| Pass criterion | Actual result equals expected result with no unexpected error |

### 8.2 Test Case Execution Table

| ID   | Feature / requirement                      | Test procedure                                                                                             | Test data                                             | Expected result                                                             | Actual result                                    | Evidence | Status |
|------|--------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|----------|--------|
| TC-1 | Customer validation (FR-01, NFR-01)        | Open Customers → New. Fill all required fields and enter a phone number that is not 10 digits. Click Save. | Phone: 98765 (5 digits); also 98765432101 (11 digits) | Save is blocked; error under Phone: Phone number must be exactly 10 digits. | Save blocked; error displayed on the Phone field | SS-1     | Pass   |
| TC-2 | Shipment weight validation (FR-02, NFR-01) | Open Shipments → New. Fill all required fields and set                                                     | Weight: 0; also -5                                    | Save is blocked; error under Weight: Shipment weight                        | Save blocked; error displayed on the Weight      | SS-2     | Pass   |

| ID   | Feature / requirement    | Test procedure                                                                                                                                        | Test data                                                 | Expected result                                                                    | Actual result                                               | Evidence | Status |
|------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------|----------|--------|
|      |                          | Weight to 0 or a negative number. Click Save.                                                                                                         |                                                           | must be greater than 0.                                                            | field                                                       |          |        |
| TC-3 | Flow automation (FR-04)  | Open a shipment with Status Booked. Create a Delivery Update for it with Status In Transit and a current Update Date. Save, then reopen the shipment. | Shipment Status before: Booked; Update Status: In Transit | Parent shipment Status changes from Booked to In Transit automatically.            | Shipment Status shows In Transit after the update was saved | SS-3     | Pass   |
| TC-4 | Invoice creation (FR-05) | Open Invoices → New. Select shipment SHIP-0001, enter amount, invoice date and payment status. Save and open the shipment.                            | Shipment: SHIP-0001; Amount and Payment Status entered    | Invoice saves with an INV number; it appears in SHIP-0001's Invoices related list. | Invoice created and linked to SHIP-0001                     | SS-4     | Pass   |
| TC-5 | Analytics dashboard      | Open Dashboard                                                                                                                                        | Data from the                                             | Donut chart                                                                        | All three components                                        | SS-5     | Pass   |

| ID | Feature / requirement | Test procedure                                    | Test data                                     | Expected result                                                                           | Actual result                  | Evidence | Status |
|----|-----------------------|---------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------|----------|--------|
|    | (FR-06, NFR-04)       | ds → Courier Management Dashboard. Click Refresh. | shipments , agents and invoices created above | shows shipment status; bar chart shows agent performance; metric shows revenue by branch. | nts rendered with current data |          |        |

The test shipment is SHIP-0001. The screenshots show the Tracking Number auto-number displaying as SHIP-{0000} in the built org, while Section 4 lists the base document's SHP-{00000}; the built format is the one that applies.

### 8.3 Screenshot Verification Artifacts

| Artifact | Test | What the screenshot must show                                                                                                                                |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SS-1     | TC-1 | New Customer form with the invalid phone entered and the red error text on the Phone field                                                                   |
| SS-2     | TC-2 | New Shipment form with Weight set to 0 or below and the error text on the Weight field                                                                       |
| SS-3     | TC-3 | Delivery Update record (UPD number, In Transit) and the parent shipment showing Status In Transit, ideally with the earlier Booked state in a second capture |
| SS-4     | TC-4 | Invoice record with INV number, plus the Invoices related list on SHIP-0001                                                                                  |
| SS-5     | TC-5 | Courier Management Dashboard with all three components populated and the refresh timestamp visible                                                           |

### 8.3.1 Screenshot Evidence

**SS-1 (TC-1): Customer validation.** A phone number of 12345 is rejected with the message Phone number must be exactly 10 digits. The save panel lists Phone (and Pincode) as fields to review.

The screenshot shows a 'New Customer' form in a web application. The form is divided into two columns. The left column contains fields for 'Customer Name' (Alex John), 'Email' (pmdfaiz07@gmail.com), 'Phone' (12345), 'Customer Type' (Regular), and 'Is Active' (checked). The right column contains fields for 'Owner' (P. MOHAMMED FAIZ), 'Address' (123 Main Street), 'City' (Ambur), and 'State' (Tamil Nadu). The 'Phone' field is highlighted with a red border and a red error message: 'Phone number must be exactly 10 digits.' A modal dialog box is open over the form, displaying the message 'We hit a snag.' and listing the fields to review: 'Pincode' and 'Phone'. The dialog box also has 'Cancel', 'Save & New', and 'Save' buttons.

**SS-2 (TC-2): Shipment weight validation.** A weight of -2.00 is rejected on the Shipment form, and the save panel lists Weight as the field to fix.

Information

\* = Required Information

Tracking Number

Owner P MOHAMMED FAIZ

\*Customer Maaz M G

\*Expected Date 9/29/2026

Delivery Agent Search Delivery Agents...

\*Weight -2.00  
Weight must be greater than zero.

Status --None--

\*Price \$250.00

Branch Search Branches...

\*Shipment Type Domestic

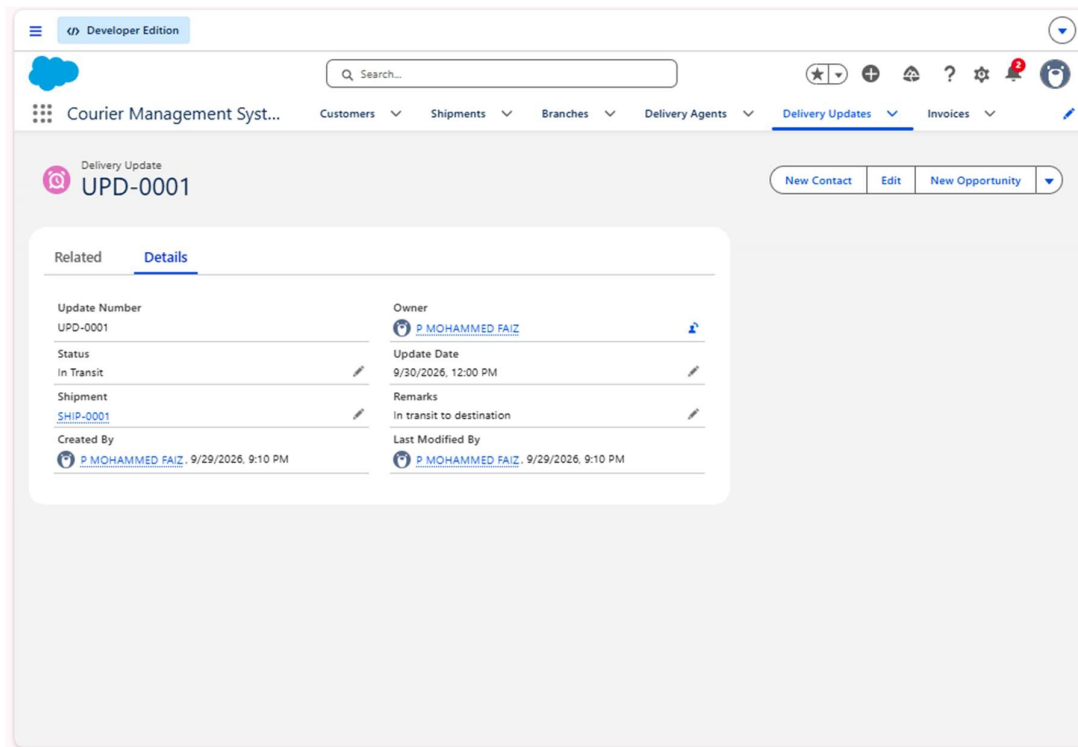
\*Source Address Chennai

\*Destination Address

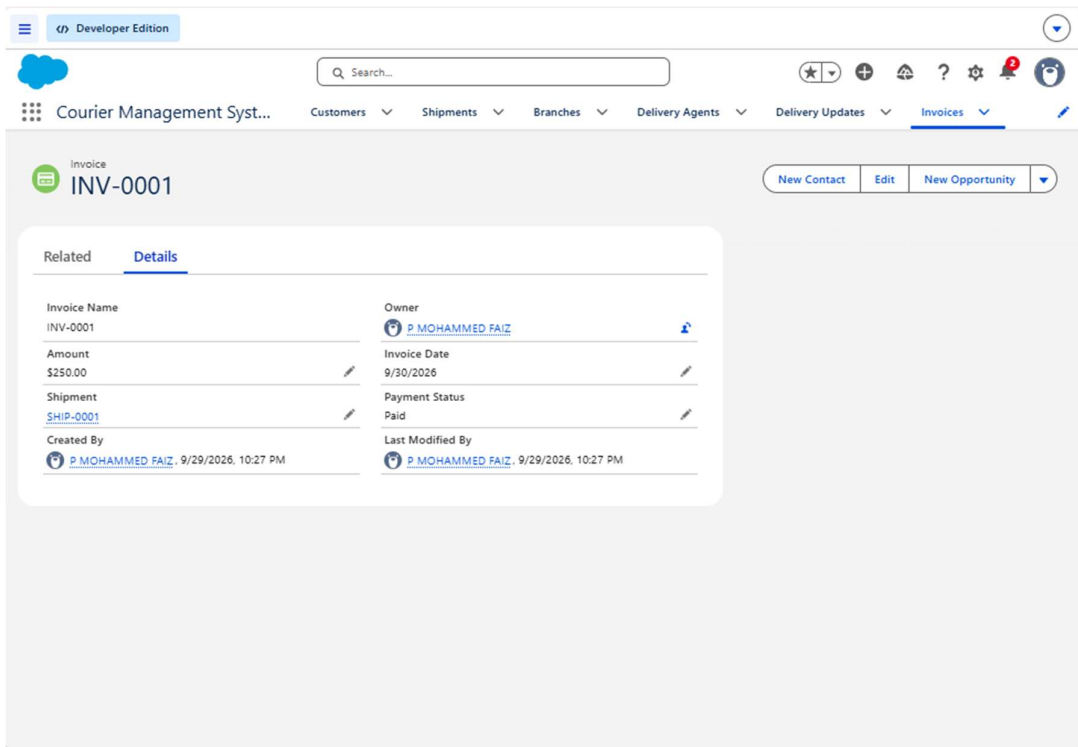
We hit a snag.  
Review the following fields  
• Weight

Cancel Save & New Save

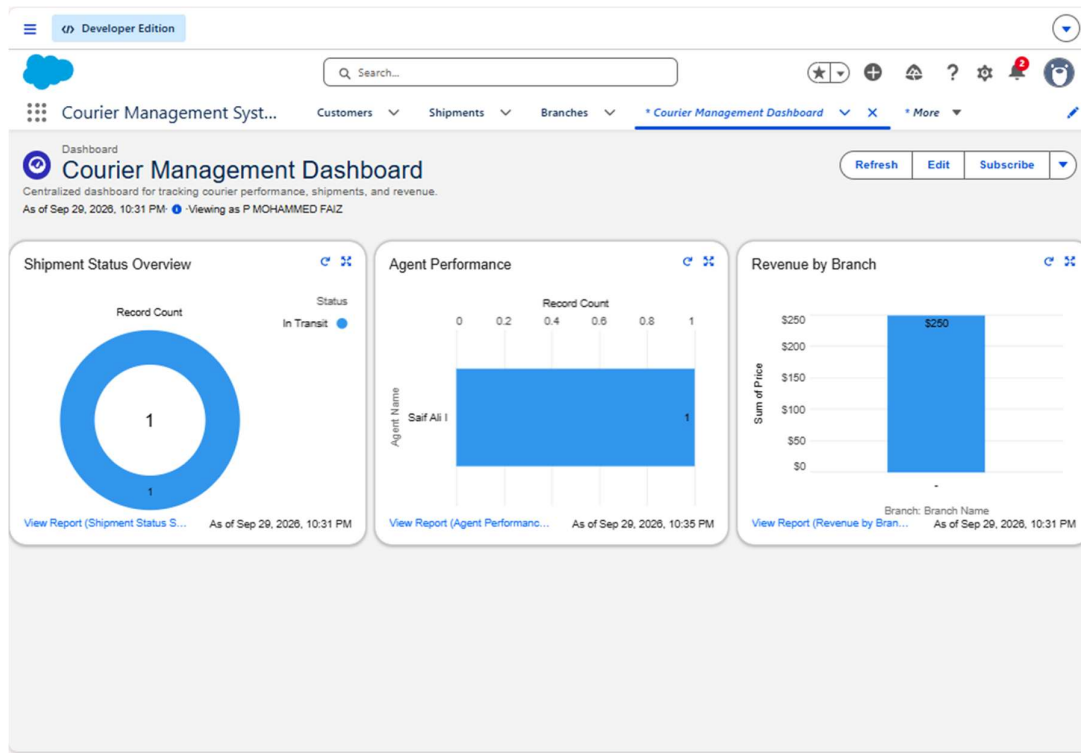
**SS-3 (TC-3): Delivery update record.** Delivery Update UPD-0001 is saved with status In Transit against shipment SHIP-0001 and the remark In transit to destination. A capture of SHIP-0001 itself showing Status In Transit would complete this test.



**SS-4 (TC-4): Invoice creation.** Invoice INV-0001 for \$250.00, dated 9/30/2026 with Payment Status Paid, is linked to shipment SHIP-0001.



**SS-5 (TC-5): Analytics dashboard.** The Courier Management Dashboard renders Shipment Status Overview (donut, 1 shipment In Transit), Agent Performance (bar chart, 1 shipment for the listed agent) and Revenue by Branch (column chart, sum of Price \$250).



#### 8.4 Additional Regression Checks Recommended

| Check                                                 | Expected result                                        |
|-------------------------------------------------------|--------------------------------------------------------|
| Customer phone with exactly 10 digits                 | Saves successfully                                     |
| Shipment weight of 0.5                                | Saves successfully                                     |
| Delivery Update with status Delayed and blank Remarks | Blocked: Remarks are required when status is Delayed.  |
| Delivery Update with status Out for Delivery          | Saves; parent shipment status not changed by the flow  |
| Shipment with expected date in the past               | Blocked: Expected delivery date cannot be in the past. |
| Branch with blank Branch Code                         | Blocked: Branch Code is mandatory.                     |
| Delivery Agent email without @                        | Blocked: Enter a valid email address.                  |
| Login as Delivery Agent, try to delete a shipment     | Delete option not available                            |
| Login as Delivery Agent, open Branches tab            | Tab not visible                                        |

#### 8.5 Test Summary

| Metric              | Value |
|---------------------|-------|
| Test cases executed | 5     |

| Metric                                             | Value                                             |
|----------------------------------------------------|---------------------------------------------------|
| Passed                                             | 5                                                 |
| Failed                                             | 0                                                 |
| Requirements covered                               | FR-01, FR-02, FR-04, FR-05, FR-06, NFR-01, NFR-04 |
| Requirements verified through configuration review | FR-03, NFR-02, NFR-03                             |

## 9. Analytics & Reporting Framework

Three grouped reports feed one dashboard, so managers can see shipment status, agent workload and branch revenue on a single page. Reports and dashboards are stored in the CMS Reports folder and shared with Courier Admin, Branch Managers and the Customer Support team.

### 9.1 Custom Report Types

| Report type label        | Primary object    | Category | Deployment status | Used by                                           |
|--------------------------|-------------------|----------|-------------------|---------------------------------------------------|
| Shipment Tracking Report | Shipment__c       | Reports  | Deployed          | Shipment Status Summary, Revenue by Branch Report |
| Customer Summary Report  | Customer__c       | Reports  | Deployed          | Customer listings and activity                    |
| Agent Performance Report | Delivery_Agent__c | Reports  | Deployed          | Agent Performance Report                          |

Create each from Setup → Report Types → New Custom Report Type: choose the primary object, enter the label, set the category, set the status to Deployed, then click Next and Save.

### 9.2 Underlying Reports

| Report                  | Report type              | Columns                                                          | Filters                                                   | Grouping and summary                           | Folder      |
|-------------------------|--------------------------|------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------|-------------|
| Shipment Status Summary | Shipment Tracking Report | Tracking Number, Customer, Status, Expected Date, Assigned Agent | Show Me: All Shipments; Date: This Month; Status optional | Group rows by Status__c; summarize Price (Sum) | CMS Reports |
| Agent Performance       | Agent Performance        | Agent Name, Branch,                                              | None by default                                           | Group by Delivery                              | CMS Reports |

| Report                   | Report type              | Columns                                          | Filters         | Grouping and summary                                                    | Folder      |
|--------------------------|--------------------------|--------------------------------------------------|-----------------|-------------------------------------------------------------------------|-------------|
| Report                   | Report                   | Number of Shipments, Status                      |                 | Agent; record count per agent                                           |             |
| Revenue by Branch Report | Shipment Tracking Report | Tracking Number, Customer, Price, Invoice Status | None by default | Group by Branch, then by Month; summarize Price or invoice Amount (Sum) | CMS Reports |

Shipment has no direct Branch field. Branch is reached through Assigned Agent → Branch, so the Revenue by Branch Report needs the Delivery Agent's Branch available in its report type, and Invoice Status needs Invoice as a related object of the report type. If the type does not expose them, add Delivery Agent and Invoice to the Shipment Tracking Report layout or build the revenue report from Invoice\_\_c with Shipment as its parent.

### 9.3 Dashboard Definition

| Property     | Value                                                                                 |
|--------------|---------------------------------------------------------------------------------------|
| Name         | Courier Management Dashboard                                                          |
| Folder       | CMS Reports                                                                           |
| Components   | 3                                                                                     |
| Shared with  | Courier Admin, Branch Managers, Customer Support Team                                 |
| Refresh      | Manual refresh, or a scheduled refresh if preferred                                   |
| Running user | Set to a user whose access covers all branches so every viewer sees consistent totals |

### 9.4 Dashboard Component Specifications

| Component | Name                         | Source report           | Visual type | Measure      | Grouping                                                            | Business question answered                |
|-----------|------------------------------|-------------------------|-------------|--------------|---------------------------------------------------------------------|-------------------------------------------|
| 1         | Shipment Status Distribution | Shipment Status Summary | Donut Chart | Record Count | Status (Booked, In Transit, Out for Delivery, Delivered, Cancelled) | What share of shipments is in each stage? |

| Component | Name                     | Source report            | Visual type | Measure                   | Grouping       | Business question answered                  |
|-----------|--------------------------|--------------------------|-------------|---------------------------|----------------|---------------------------------------------|
| 2         | Agent Performance Matrix | Agent Performance Report | Bar Chart   | Record Count of shipments | Delivery Agent | Which agents carry the most shipments?      |
| 3         | Revenue by Branch        | Revenue by Branch Report | Metric      | Sum of Price or Amount    | Branch         | How much revenue does each branch generate? |

Build steps for each component: on the dashboard, click + Component, select the source report, choose the visual type, set the measure and grouping in the component editor, and click Add. Metric components display a single headline value; to compare several branches at a glance, add a metric per branch or switch Component 3 to a column chart, as the base build did.

## 9.5 Dashboard Layout

|                                         |                                   |
|-----------------------------------------|-----------------------------------|
| Courier Management Dashboard            |                                   |
| 1. Shipment Status Distribution (Donut) | 2. Agent Performance Matrix (Bar) |
| 3. Revenue by Branch (Metric)           |                                   |

## 9.6 Save, Share and Refresh

1. Click Save, then Done.
2. Choose Share on the CMS Reports folder and grant Viewer access to the three groups.
3. Open the dashboard and click Refresh to load current data; the last-refreshed time appears at the top.
4. Dynamic dashboards can show each viewer their own data, but a fixed running user is used here to keep branch totals consistent.

## 10. User Guide & Deployment Manual

Branch Managers book and monitor shipments, Delivery Agents post tracking updates, and administrators deploy and maintain the configuration. Each group follows the short procedures below.

### 10.1 Branch Manager Guide

1. **Sign in.** Log in to Salesforce and open the Courier Management System app from the App Launcher.
2. **Register the customer.** Go to Customers → New. Enter name, customer type, email, a 10-digit phone, address and a 6-digit pincode. Tick Is Active and Save. Fix any red field error shown before saving again.

3. **Book the shipment.** Go to Shipments → New. Choose the customer, shipment type, both addresses, weight, price and expected delivery date. Leave Status as Booked. Save; the system assigns the tracking number.
4. **Assign an agent.** On the shipment, set Assigned Agent to an agent whose status is Active and whose branch fits the route. Save.
5. **Monitor progress.** Open the shipment and read the Delivery Updates related list. Status changes to In Transit when the first In Transit update is logged.
6. **Raise the invoice.** Open Invoices → New, choose the shipment, enter amount, invoice date and payment status. Save. Change Payment Status from Unpaid or Pending to Paid when payment is received.
7. **Review performance.** Open Dashboards → Courier Management Dashboard and click Refresh. Drill into a chart to see its source report.
8. **Maintain agents and branches.** Update agent status to On Leave or Inactive when needed, so assignment lists stay accurate.

## 10.2 Delivery Agent (Field Agent) Guide

1. Open the Shipments tab and find the parcels assigned to you.
2. At each milestone, open Delivery Updates → New. Select the shipment, choose the status (In Transit, Out for Delivery, Delivered or Delayed), and confirm the update date and time.
3. If the status is Delayed, enter the reason in Remarks; the system will not save without it.
4. Save. Open the shipment to confirm the updated status.
5. You can view customers and invoices but cannot edit branches or delete records. Ask your Branch Manager for any correction outside your access.

## 10.3 Customer Support Guide

1. Search the customer by name in Customers.
2. Open Shipments to read the current status and the Delivery Updates list.
3. Correct customer contact details when the customer reports a change.
4. Escalate delayed or disputed shipments to the Branch Manager.

## 10.4 Deployment Instructions

CMS is configuration only, so it deploys as metadata. The order below respects dependencies; an object must exist before its fields, and lookups need their target objects first.

| Step | Component                    | Notes                                                                                                                     |
|------|------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | Create Developer Edition org | <a href="https://developer.salesforce.com/signup">developer.salesforce.com/signup</a> ; verify email and set the password |
| 2    | Custom objects               | Branch, Customer, Delivery Agent, Shipment, Delivery Update, Invoice                                                      |
| 3    | Fields and relationships     | Create parents before children; Delivery Update's master-detail is best created while the object has no records           |

| Step | Component             | Notes                                                     |
|------|-----------------------|-----------------------------------------------------------|
| 4    | Validation rules      | Nine rules from Section 5                                 |
| 5    | Page layouts and tabs | Sections and related lists per Section 7                  |
| 6    | Flow                  | Build and activate Update_Shipment_Status_Flow            |
| 7    | Roles and profiles    | Roles first, then profiles and tab and object permissions |
| 8    | Lightning app         | Add all navigation items and all four profiles            |
| 9    | Users                 | Create one user per group; match license to profile       |
| 10   | Reports and dashboard | Report types, reports, dashboard, folder sharing          |

To move the configuration to another org, retrieve the metadata with the Salesforce CLI and deploy it to the target org, or build an unlocked package; developer orgs cannot send outbound change sets to production. Deploy the flow as Active only after the objects, fields and permissions exist in the target.

### 10.5 Post-Deployment Verification Checklist

- ☐ All six objects, tabs and page layouts are visible in the Courier Management System app
- ☐ Customer with a phone number that is not 10 digits is blocked
- ☐ Shipment with weight 0 is blocked
- ☐ Delayed update without remarks is blocked
- ☐ A new In Transit Delivery Update changes the parent shipment from Booked to In Transit
- ☐ Update\_Shipment\_Status\_Flow shows status Active
- ☐ Invoice links to its shipment and appears in the related list
- ☐ Each of the four users can sign in and sees only the tabs listed for the profile
- ☐ Delivery Agent cannot delete records or edit branches
- ☐ Reports run without error; Courier Management Dashboard refreshes with all three components
- ☐ CMS Reports folder is shared with Courier Admin, Branch Managers and Customer Support
- ☐ Sample data used for testing is removed or clearly labeled

### 10.6 System Maintenance Recommendations

| Area         | Recommendation                                          | Frequency |
|--------------|---------------------------------------------------------|-----------|
| Users        | Deactivate leavers; review profile and role assignments | Monthly   |
| Data quality | Review shipments stuck in                               | Weekly    |

| Area                | Recommendation                                                                                                                                                                            | Frequency          |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                     | Booked or In Transit beyond expected date                                                                                                                                                 |                    |
| Flow                | Extend Update_Shipment_Status_Flow with a Decision element covering Out for Delivery and Delivered; add Delayed to the Shipment Status picklist only if the business wants it shown there | Next release       |
| Validation          | Adjust the expected-date rule so it applies to new records or date changes only                                                                                                           | Next release       |
| Reports             | Confirm branch grouping still works after adding branches; archive unused reports                                                                                                         | Quarterly          |
| Backup              | Run Data Export from Setup to keep an offline copy                                                                                                                                        | Monthly            |
| Salesforce releases | Review release notes for flow and report changes                                                                                                                                          | Three times a year |

Future scope, when demand justifies it: mobile tracking app, GPS integration, SMS notifications and AI-based delivery optimization.

### 10.7 Advantages and Limitations

| Advantages          | Limitations                                      |
|---------------------|--------------------------------------------------|
| Centralized data    | Dependence on Salesforce                         |
| Automated workflows | Initial configuration effort                     |
| Real-time reporting | Developer Edition limits on licenses and storage |

### 10.8 Source Reconciliation Notes

The report follows the specification you supplied. Where the base Word document differs, both versions are recorded here so the team can pick one.

| Topic            | Base Word document                                            | This report                                                                            |
|------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Phone validation | LEN(Phone__c) <> 10, message: Phone number must be 10 digits. | NOT(REGEX(Phone__c, "^[0-9]{10}\$")), message: Phone number must be exactly 10 digits. |
| Weight message   | Weight must be greater than                                   | Shipment weight must be                                                                |

| Topic                | Base Word document                                                                                       | This report                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                      | zero.                                                                                                    | greater than 0.                                                                                                              |
| Flow name and scope  | Auto_Update_Shipment_Status;<br>runs on create with no<br>condition and copies Status to<br>the shipment | Update_Shipment_Status_Flow;<br>runs on create or update when<br>Status is In Transit and sets the<br>shipment to In Transit |
| Revenue component    | Column chart                                                                                             | Metric component                                                                                                             |
| Test shipment label  | Auto-number SHP-{00000}                                                                                  | Test brief label SHIP-0001                                                                                                   |
| Sprint tracker dates | Sprint dates shown as 2022                                                                               | Not reproduced; update to the<br>actual 2026 sprint dates before<br>submission                                               |
| Milestone 1 text     | Refers to a Loan Management<br>System                                                                    | Not carried over; the<br>introduction belongs to a<br>different project template                                             |